Memory (Group 4)

Author roles

Author roles were classified using the Contributor Role Taxonomy (CRediT; credit.niso.org) as follows:

Martina Iscar: project administration, conceptualization, data curation, formal analysis, methodology, investigation, software, validation and writing review and editing.

Irene Morales: project administration, conceptualization, data curation, software, formal analysis, visualization writing review and editing.

Amalia Varo: project administration, conceptualization, data curation, software, formal analysis, formal analysis, methodology, visualization writing review and editing.

Ferrán Extremera: Project administration, conceptualization, data curation, methodology, formal analysis, software, validation visualization writing review and editing.

Use of Generative AI in This Assignment

This project used generative AI as a collaborative tool. Specifically, we employed ChatGPT and Gemini to develop the R code by giving it specified prompts, clarify statistical interpretations regarding regression outputs, and improve the storytelling flow of our HR recommendations.

Even though these tools provided the foundation for the code, human intervention was present at every stage. We frequently modified and corrected overcomplicated code and addressed false interpretations—particularly regarding the counter-intuitive relationship between job strain and performance. We understood the nuances of the manufacturing and financial databases much more clearly than the AI. Therefore, we were the primary decision-makers, remaining responsible for the content, explanations, and strategic conclusions. AI was a tool for enhancement, not a replacement for our critical thinking.

Critical Reflection:

Our experience from previous cases using AI made us more aware of the deficiencies and what strategy works best. However, in this case to get a more uniform code, one person developed a basic code during class with AI, and then we continued with our section approach were each of the group members had specific prompts to work on and improve. Adding the analysis and interpretations required to each of the deliverables.

In addition, similarly to case 3, to improve our efficiency for Case 4, we shifted our strategy: instead of just asking for code, we first asked the AI to explain the **"Theoretical Link"** between health (sick days), attitudes (job satisfaction), and performance. This allowed us to anticipate the results before running the models in R. This help develop good initial codes.

We also took a more active role in "prompt engineering" to ensure the AI didn't confuse **Individual Performance** (Prompts 4-6) with **Customer Loyalty** (Prompts 1-3).

A significant challenge occurred during Prompt 5 (Stepwise Regression). The AI initially suggested a "Forward" selection, but based on what we learned in class, we corrected it to use a "Both" (bidirectional) approach to ensure a more robust model. We also had to manually interpret the "Job Strain" paradox, as the AI initially assumed strain would always decrease performance, while our data showed a positive correlation in this specific manufacturing context.

Finally, we used AI as a translator and tone-checker. Some of our strategic recommendations were drafted in Spanish to ensure the HR logic was sound, then translated into professional English, ensuring the delivery was appropriate for an executive-level report.

Overall, AI did provide powerful insights and helped us deepen our understanding of the case. Nevertheless, it is our responsibility to proofread the answers given and correct where necessary.

Examples of Significant AI Prompt–Response Interactions

Example 1: Identifying Appraisal Bias (Prompt 8)

- **Prompt:** Predict performance using Salary, Gender, and Country. If 'Country' is significant, explain from an HR perspective if this means people in certain countries are 'better' workers or if there is a problem with the appraisal system.
- **Response:** The AI pointed out that if Country dummies are significant after controlling for salary, it likely indicates a Subjective Bias in how managers in different regions rate their staff, rather than a difference in actual ability.
- **Evaluation:** This interpretation was vital. It allowed us to recommend a "Calibration" program for managers to ensure that a "4/5 rating" means the same thing in every country, preventing legal and fairness issues.

Example 2: Analysing Sickness Absence Limitations (Prompt 7)

- **Prompt:** Predict SickDays2014 using attitude variables. If the results are non-significant, explain why this happens and what other data HR should collect to better understand absenteeism.
- **Response:** It showed that attitudes like "Job Satisfaction" had almost zero impact on sick days. It suggested that sickness is often "Stochastic" (random) or driven by physical health data which HR doesn't usually have for privacy reasons.
- **Evaluation:** This taught us that HR shouldn't blame "low morale" for every sick day. We adjusted our recommendation to suggest "Wellness and Preventive Health" programs instead of just "Engagement" surveys to solve absenteeism.

Example 3: Linear vs. Non-Linear Patterns (Prompt 9)

- **Prompt:** In the supermarket dataset, the linear model for Age isn't showing much. Help me write a model using the age2 through age10 dummy variables. Compare the R^2 and tell me which age group is actually the fastest at scanning.
- **Response:** It provided the code comparing the two models. It showed that the R^2 jumped from 0.24 to 0.38, revealing a "hump-shaped" productivity curve where middle-aged workers (age 4-6) are the peak performers.
- **Evaluation:** We caught a small error where the AI tried to include all 10 age variables; we corrected the code to leave age1 out as the reference group to avoid the "dummy variable trap."

Example 4: Connecting Metrics and the Evolution of Predictive Power

- **Prompt:** Review the R^2 values from Prompt 1 (33%), Prompt 4 (9%), and Prompt 9 (38%). Explain why HR data can predict Customer Loyalty and Checkout Speed so well but struggles to predict "Performance Ratings" in manufacturing.
- **Response:** The AI provided a deep dive into "Objective vs. Subjective" metrics. It explained that Checkout Speed (Prompt 9) is a "Hard Metric" (easy to predict), and Customer Loyalty (Prompt 1) is a "Direct Response" to a person. However, Performance Ratings (Prompt 4) are "Subjective" and often influenced by factors the survey doesn't measure, like technical skill or manager-employee relationships.
- **Evaluation:** This helped us understand the **Evolution of Analytics**. We learned that HR has the most impact when they measure "Hard" behaviours (scanning speed) or "Immediate" feelings (loyalty), but they need more

complex models (like Prompt 6, adding baseline data) to understand year-to-year performance growth. This comparison gave our report a much more professional, analytical tone.

Example: Demographic Bias in Performance Ratings (Prompt 8)

- **Prompt:** In this employee performance dataset, can you help me build a regression model using SalaryBenchmark, Gender, AgeCat, TenureCat, and Country, and then explain whether gender and country have a significant impact on performance ratings and if there are signs of bias?
- **Response:** The regression results showed that salary has a positive effect on performance, while tenure also increases ratings. However, gender has a strong and significant negative effect, and country variables show large differences across locations. These findings suggest that performance ratings are influenced not only by objective factors but also by demographic characteristics, indicating potential bias.
- **Evaluation:** The model correctly handled categorical variables by using factors and avoiding the dummy variable trap. The interpretation appropriately focused on the significant effects of gender and country as indicators of potential bias and inconsistencies in the appraisal system.